European AI Challenge 2026: From Data to Policy

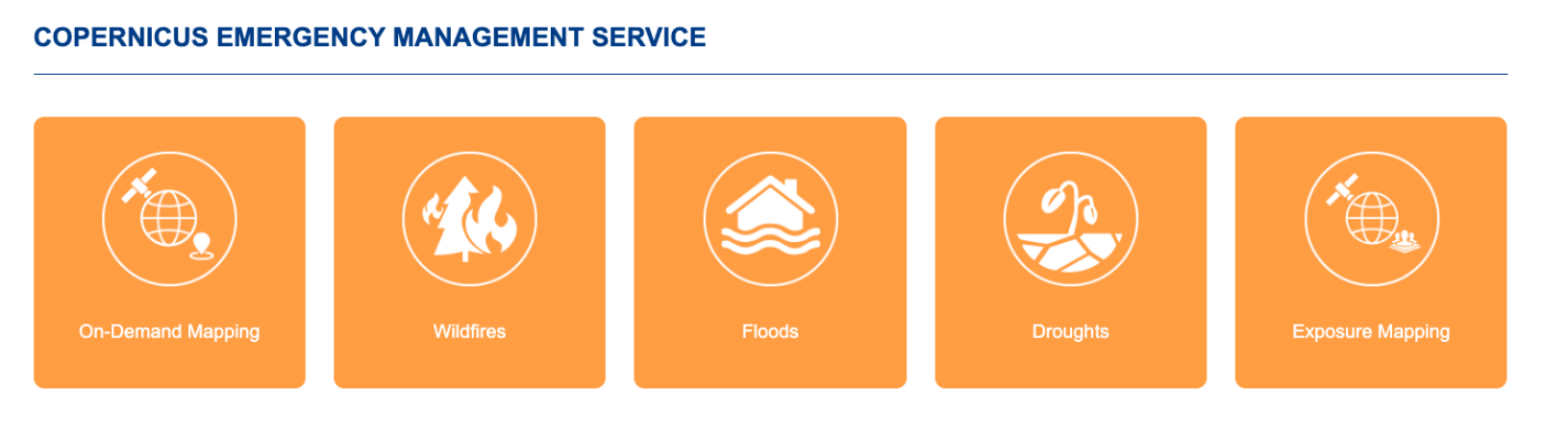
Presentation of the datasets

Martin Jančovič, DG JRC E1, Disaster Risk Management - External consultant for the European commission (KAJO)
Thomas Saillour, DG JRC E1, Disaster Risk Management - Coastal Floods
Olesya Grabova, DG JRC S4, Science for Policy

European Commission

Copernicus Emergency Management Service (CEMS)

https://emergency.copernicus.eu/



Floods
EFAS European Flood Awareness System

Flood hazard maps

https://data.jrc.ec.europa.eu/collection/id-0054

CEMS-EFAS flood hazard

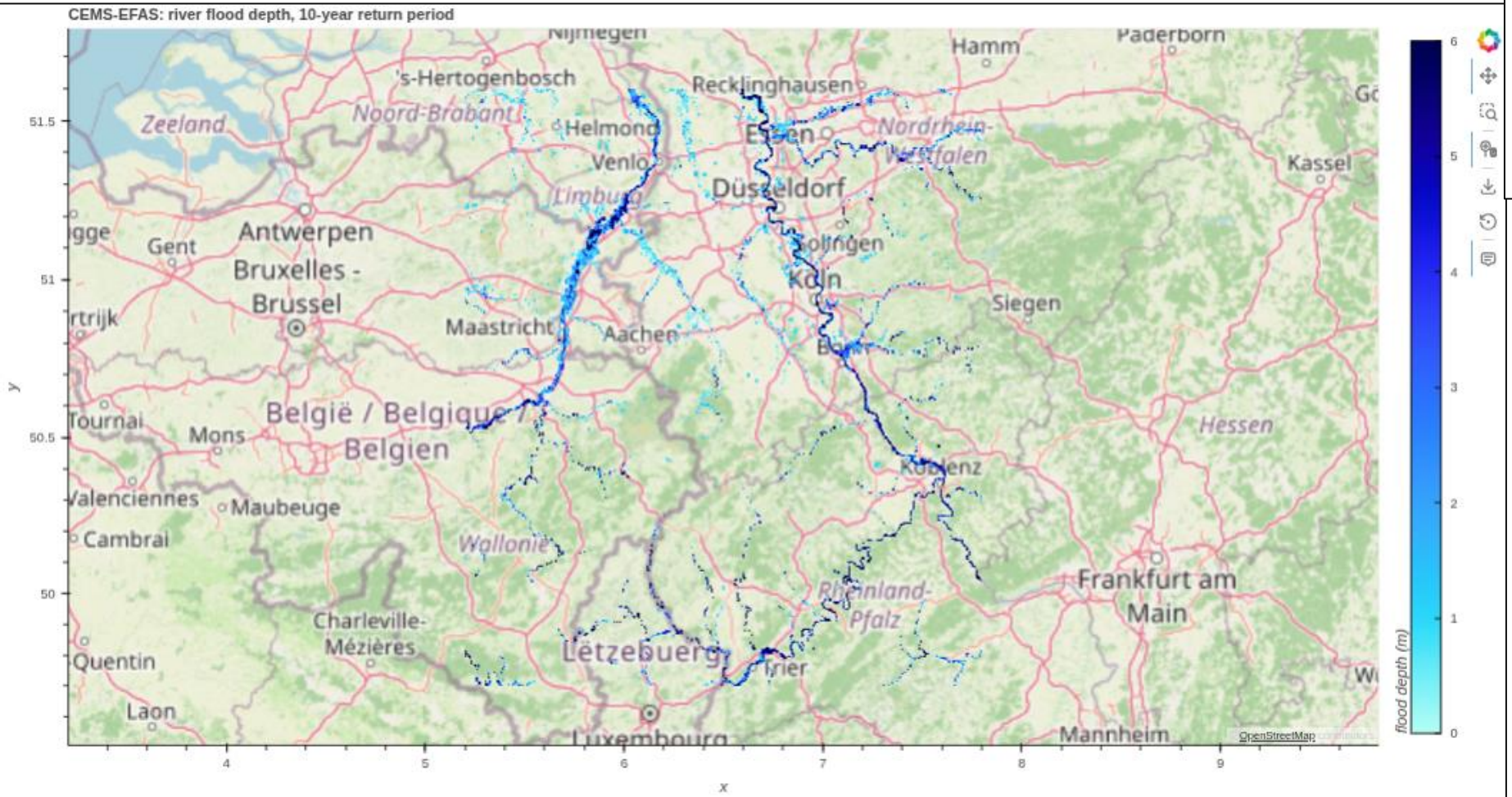
Google Cloud folder	cems-efas
Dataset	EFAS river flood hazard maps for Europe. Resolution 3 arc-second (~90 m) grid
What it is	Model-based maps of how deep the water gets in a 10-year, 100-year and 500-year river flood, for the whole of Europe.
Answers	Where does a model expect water in a 10-, 100-, 500-year river flood?
Data type	Raster (water depth in meters)
Format and size	3 GeoTIFFs, 0.95 GB in total RP10, RP100, RP500

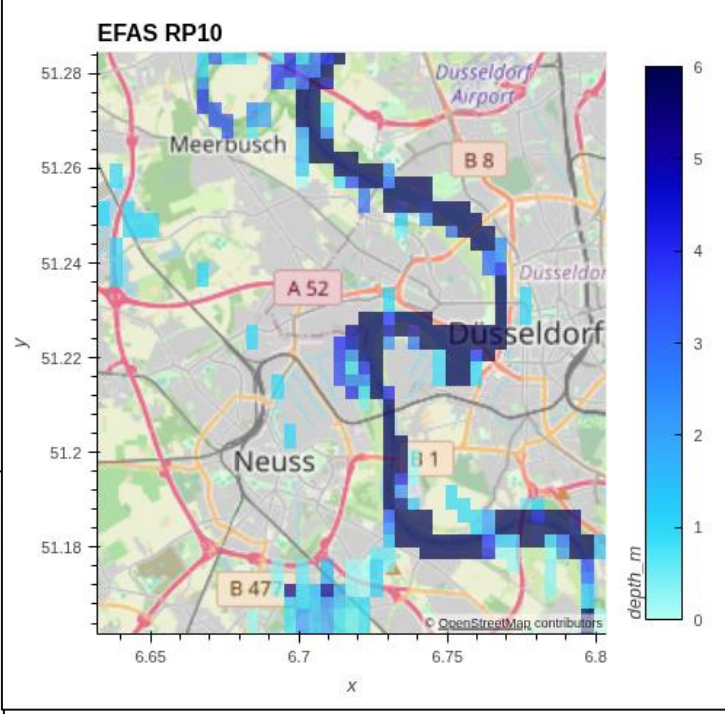
Return Periods?

- Recurrence interval = (n+1) / m
  - *n* number of years on record;
  - *m* is the number of observed occurrences
- The theoretical return period between occurrences is the inverse of the average frequency of occurrence
- Simple terms:
  - RP500 **X** an event that occurs every 500 years
  - RP500 **=** an event that has 1/500=0.002 or 0.2% probability of occurring this year

CEMS-EFAS flood hazard

- RP10

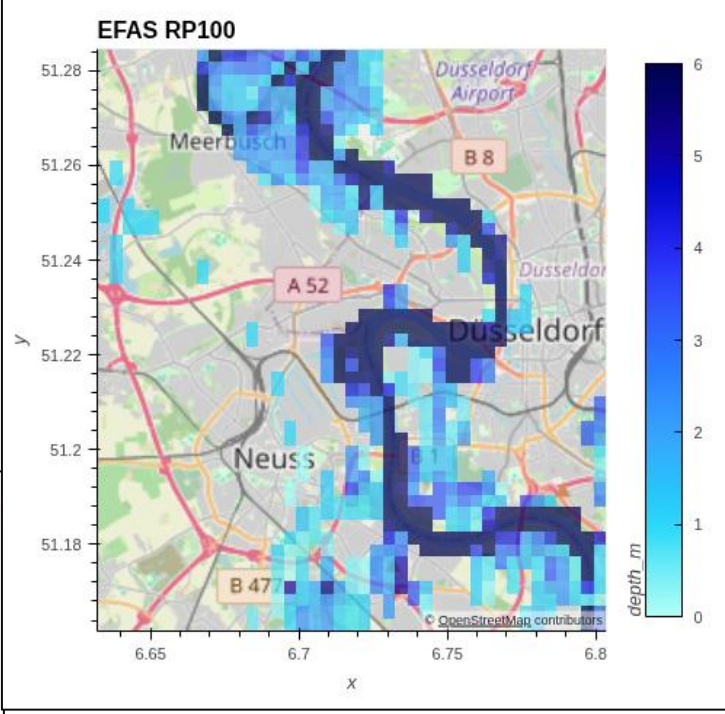




CEMS-EFAS flood hazard

- RP100

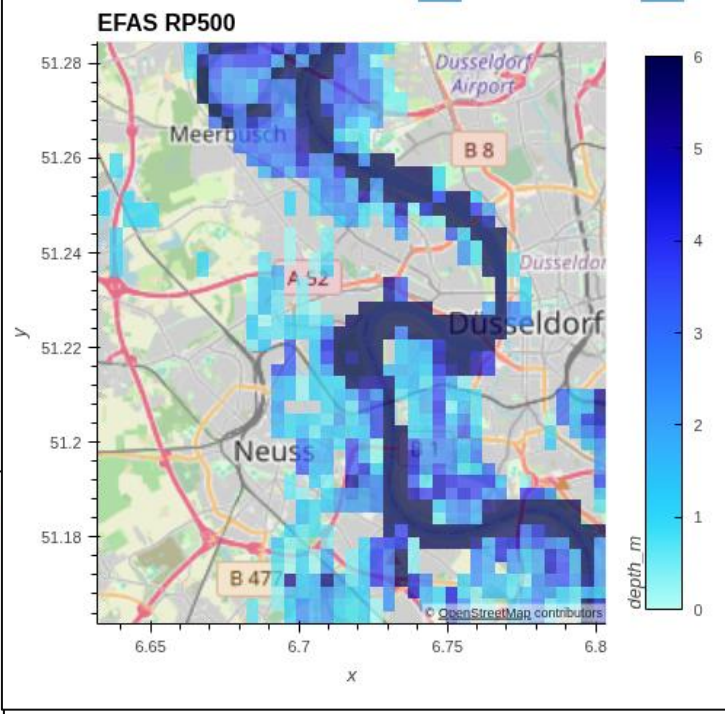




CEMS-EFAS flood hazard

- RP500





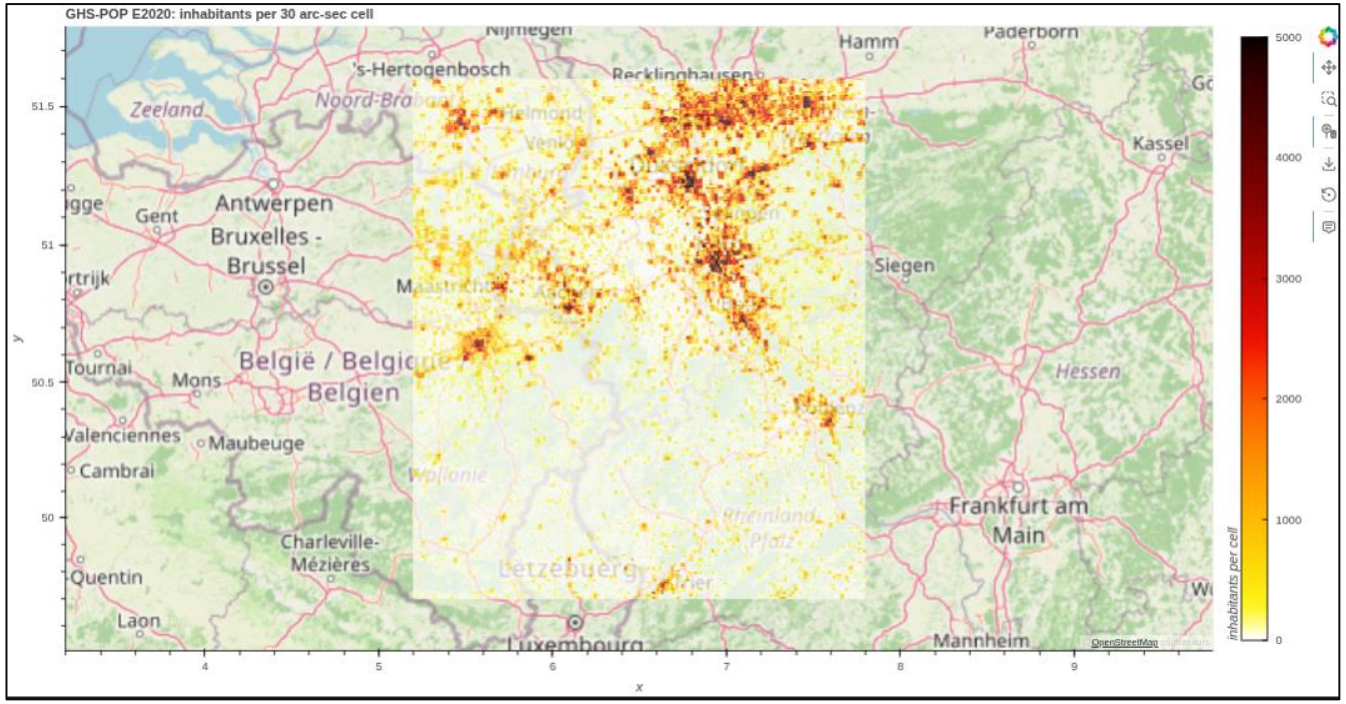
Exposure Mapping
GHSL Global Human Settlement Layer

Population Grid (POP)

https://data.jrc.ec.europa.eu/dataset/2ff68a52-5b5b-4a22-8f40-c41da8332cfe

GHS-POP

Google Cloud folder	ghs_pop
Dataset	GHS-POP R2023A
What it is	Estimated number of residents per grid cell, for the years 2020 and 2025 (projection)
Answers	How many residents per grid cell?
Data type	Raster (people per cell)
Format and size	2 GeoTIFFs of about 0.5 GB (2020 and 2025), each with an overview file (.ovr), a PDF and an Excel file



Exposure Mapping
GHSL Global Human Settlement Layer

Open Building attribute table (OBAT)

https://data.jrc.ec.europa.eu/dataset/f41a22f1-5741-4c41-86eb-6384654f6927

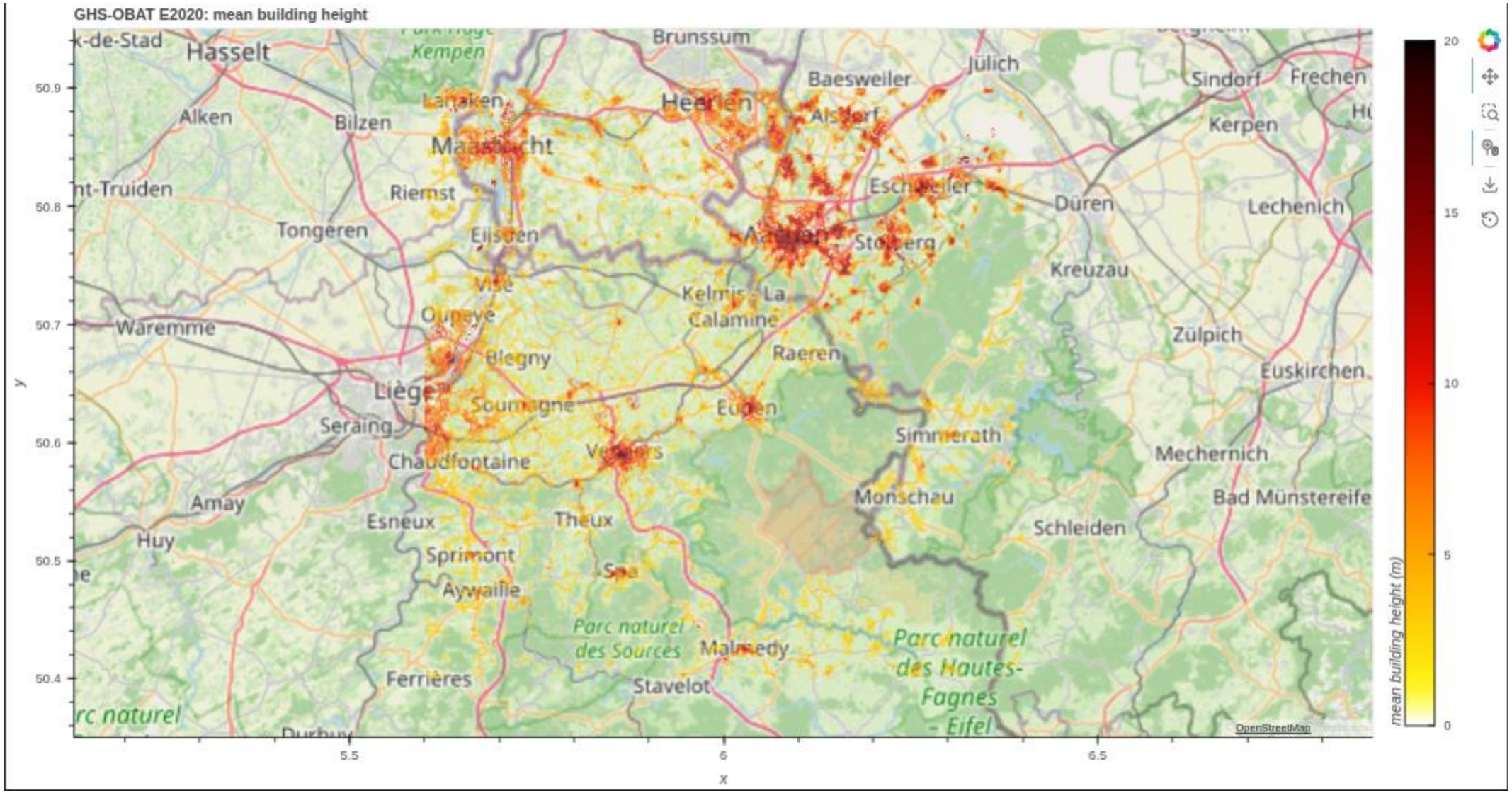
GHS-OBAT

Google Cloud folder	obat_be, obat_de, obat_nl
Dataset	GHS-OBAT R2024A
What it is	One record per building: location, use (residential or not) and, where known, height and construction period. Reference year 2020.
Answers	Which buildings were there, and are they residential?
Data type	Vector (one point per building), reference year 2020. Fields: location, use, and where known height and construction period
Format and size	One GeoPackage plus a PDF per country: BE 1.44 GB, NL 2.23 GB, DE 8.58 GB

Details

- 1 GeoPackage per country (GER, NL, BE)
  - 7.5 M points for BE
  - 44.1 M points for DE
  - 11.8 M points for NL
- Coded fields, per the GHS-OBAT R2024 documentation:
  - `**height**`:mean building height (m), from GHS-BUILT-H ANBH, lower-bounded at 2.5 m
  - `**use**`: 0 outside built-up domain, 1 residential, 2 non-residential
  - `**epoch**`: construction epoch: 0 outside, 1 `<1980`, 2 `1980-1990`, 3 `1990-2000`, 4 `2000-2010`, 5 `2010-2020`
  - `**shapefactor**`: surface-to-volume ratio (m^2/m^3), inversely proportional to building size
  - `**area**`, `perimeter`: footprint area (m^2) and perimeter (m), computed in the local UTM zone
  - `**adm1**`: GADM level-1 administrative unit

GHS-OBAT


GHS-OBAT E2020: mean building height
Hasselt
Brunssum
Kempen
Baesweiler
Jülich
Sindorf
Frechen
Alken
Bilzen
Heerlen
Kerpen
Maastricht
Riemst
Düren
Lechenich
Tongeren
Eijsden
Stolberg
Kreuzau
Kelmis - La Calamine
Waremme
Zülpich
Blegny
Raeren
Euskirchen
Liège
Soumagne
Eupen
Seraing
Simmerath
Chaudfontaine
Mechernich
Amay
Monschau
Bad Münstereifel
Esneux
Theux
Schleiden
Huy
Sprimont
Aywaille
Parc naturel des Sources
Malmedy
Parc naturel des Hautes-Fagnes - Eifel
Ferrières
Stavelot
OpenStreetMap
mean building height (m)
20
15
10
5
0
50.9
50.8
50.7
50.6
50.5
50.4
5.5
6
6.5
x
y

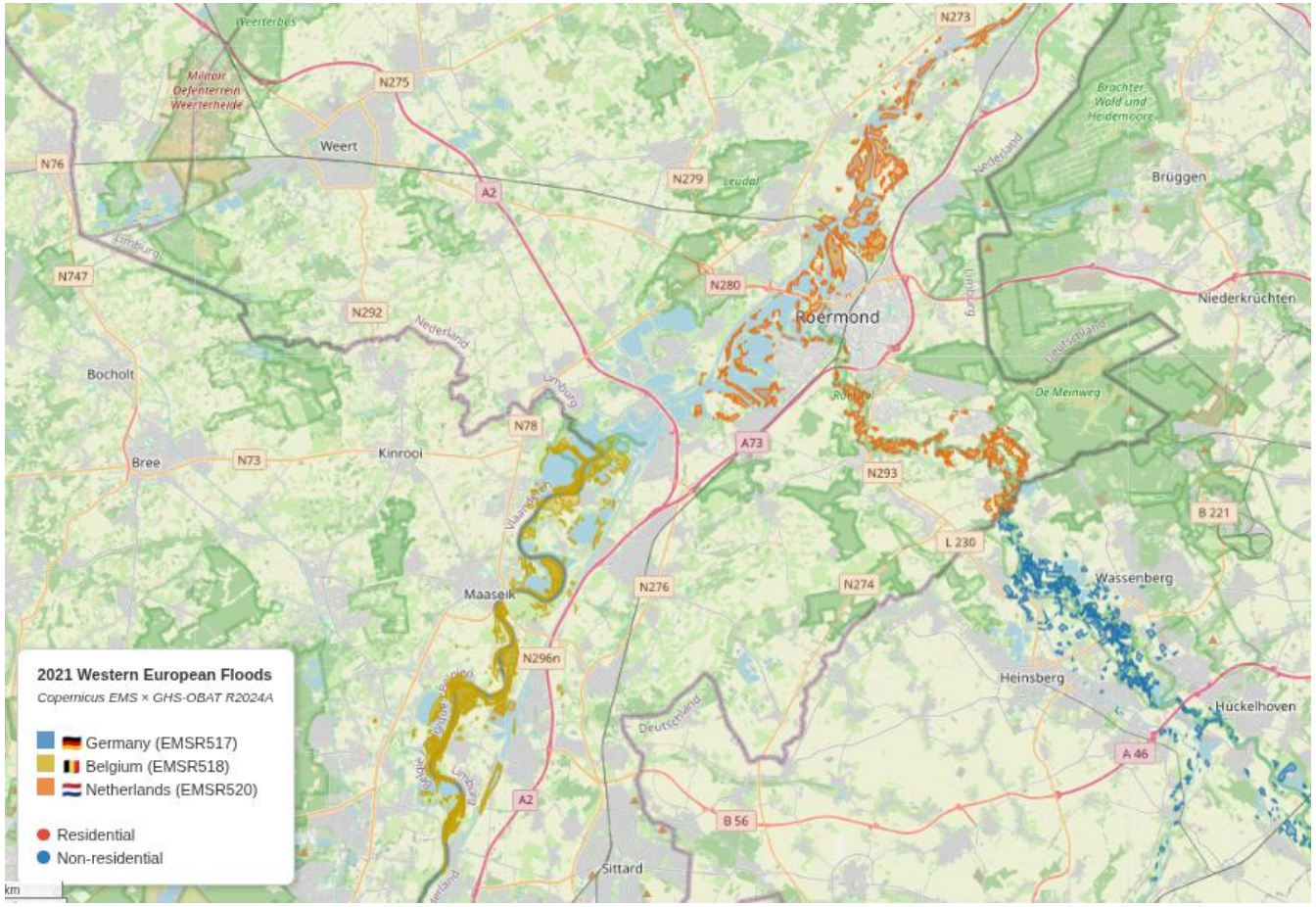
On Demand Mapping

Flood maps activations

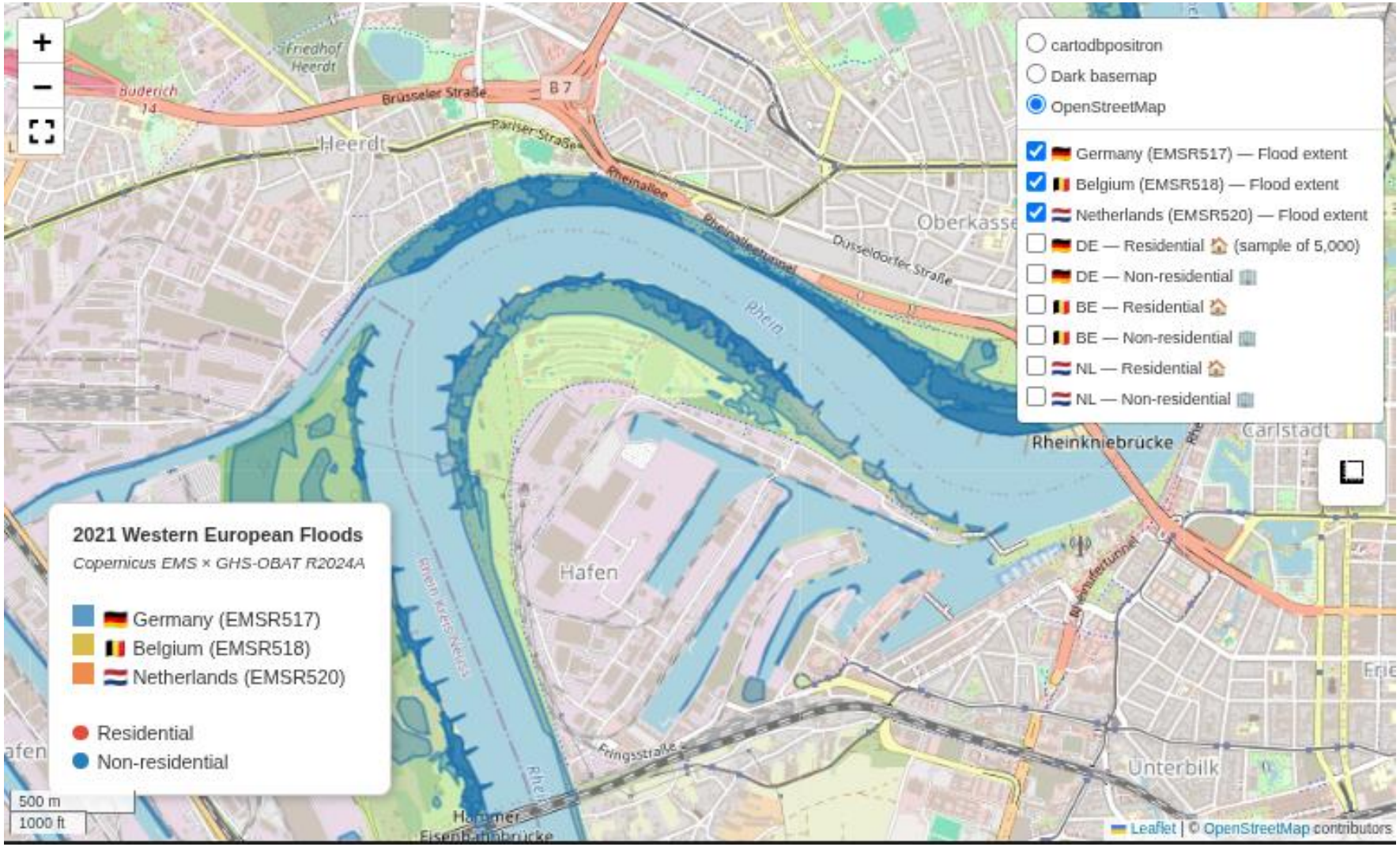
https://mapping.emergency.copernicus.eu/

Flood maps activations

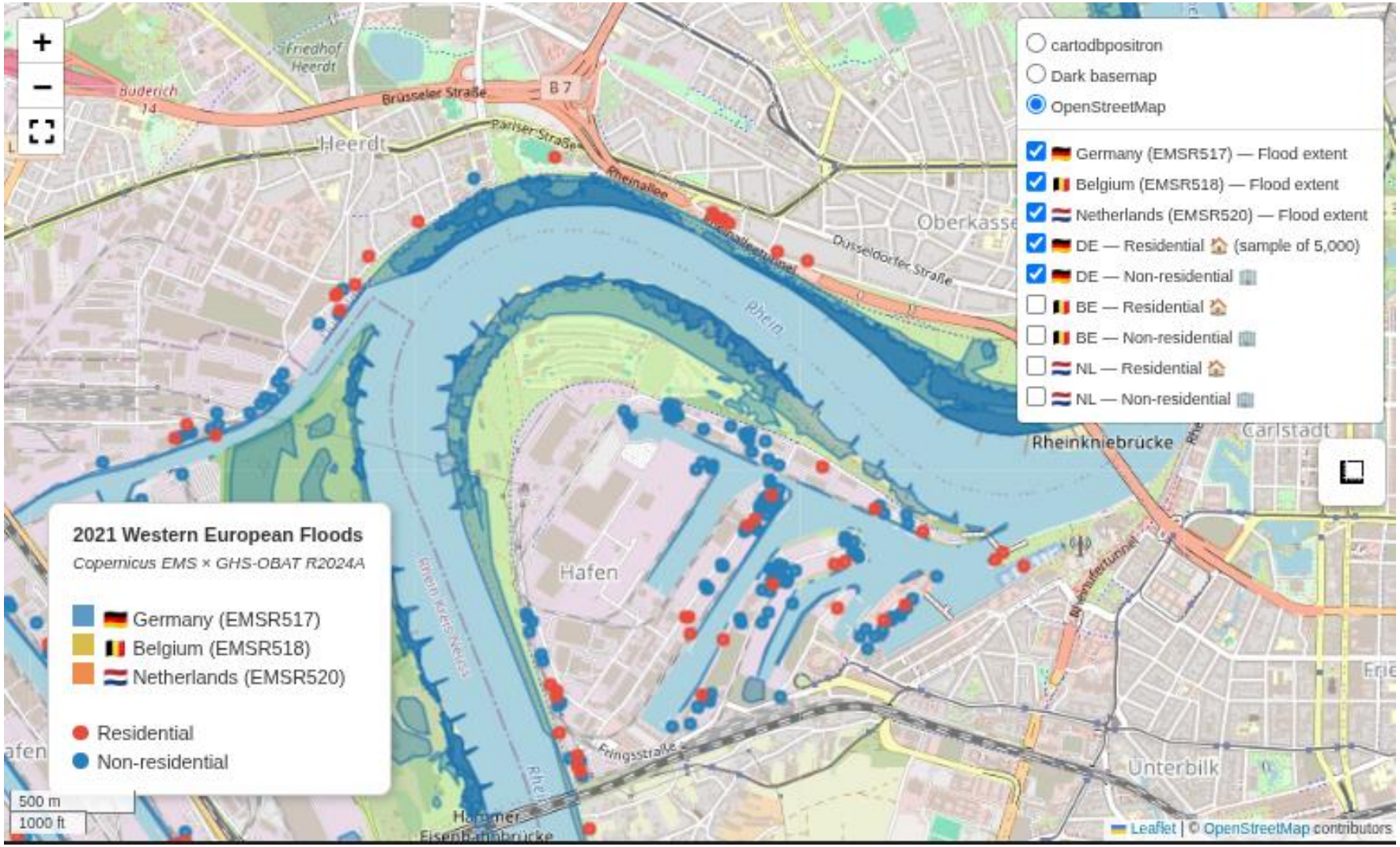
Google Cloud folder	ems_be, ems_de, ems_nl
Dataset	CEMS Rapid Mapping (activations EMSR518 BE, EMSR517 DE, EMSR520 NL)
What it is	Flood maps drawn by analysts from satellite images during the 2021 floods: where the water was and which buildings were hit.
Answers	Where was the water, as seen by satellite?
Data type	Vector: polygons, points, lines drawn by analysts from satellite images
Format and size	One folder per product, with shapefiles plus JSON, KMZ and XML copies: BE 4 products (0.39 GB), DE 22 (0.54 GB), NL 1 (0.06 GB)



Flood maps activations – zoom



Combine information



Get the slides

https://gist.github.com/MartinMIB-BA/45e8075ee8cd87e6137ff28aeeb5ed84

Get the data (Google Cloud)

https://console.cloud.google.com/storage/browser/datasets_challenge/working_data

Thank you



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